Supplementary information for "Collective Motion with Memory" by K.H.Nagai *et al.*

I. NUMERICAL METHODOLOGY FOR PARTICLE MODELS

Simulations of our continuous-time Vicsek-style models were performed on (mostly square) domains with periodic boundary conditions, using in most cases simple Euler time-stepping with typical time step 0.1.

For establishing the boundaries of the various regions in the phase diagrams (Fig. 1), the following methodology was used: a first scan of the parameter plane from several random and ordered initial conditions yielded a number of collective steady states. Each of these states was then followed adiabatically in the parameter plane starting from parameter values at which it was spontaneously observed, yielding a line delimiting their existence. The stability of the results was checked against changes in the system size and the maximum time used to declare a typical state stable (typically $T = 2 \times 10^5$).

Coarse-grained scalar fields were used to build the snapshots presented in Fig.2 and the movies. These fields were calculated by averaging in space on boxes of size 4×4, with a local time-average over $\Delta t = 10$. We thus defined the local polar intensity field $P(x, y, t)$ and local nematic intensity field $Q(x, y, t)$, as well as the corresponding local orientation fields $\Theta_P(x, y, t)$ and $\Theta_Q(x, y, t)$. Similarly, coarse-grained density fields $\rho(x, y, t)$, $\rho_{\rm CW}(x, y, t)$ and $\rho_{\rm CCW}(x, y, t)$ were defined, the last two distinguishing clockwise ($\omega > 0$) and counter-clockwise ($\omega < 0$) particles.

To define and distinguish the various phases reported in the phase diagrams, we of course used the global quantifiers:

- intensity of polar order $P_{\rm g} = |\langle \exp(i\theta_j)\rangle|$
- intensity of nematic order $Q_{\rm g} = |\langle \exp(2i\theta_j)\rangle|$.

But we also used averages of the coarse-grained fields defined above:

- average intensity of local polar order parameter $P_\ell = \langle P(x, y, t)\rangle$
- average intensity of local nematic order parameter $Q_\ell = \langle Q(x, y, t)\rangle$
- variance of local density field $R_\ell = \langle \rho(x, y, t)^2\rangle - \rho_0^2$

where the average $\langle . \rangle$ is taken over space and time.

The various phases described in the text and represented on the phase diagrams (Fig.1) are then characterized by a unique combination of finite (order 1) and near-zero values of the above quantities, as summarized in the Table I.

TABLE I. Characterization of the various phases defined in the text. For each phase, 5 scalar quantifiers were calculated. Here we mark whether they were (near) zero (0) or of order 1 (1). For some phases, additional characteristics had to be used: In the nematic model, vortex lattice (V) and active foam (AF) were distinguished by the structure of the (time-averaged) spatial Fourier spectrum. For the polar model, the spatial Fourier spectrum was also used to distinguish the vortex lattice (V) from vortex chaos (VC). The pdf of particle's headings was used to separate the Vicsek waves (VW) from the zigzag (ZZ).

nematic model

phase	$Q_{\rm g}$	$P_{\rm g}$	R_ℓ	Q_ℓ	P_ℓ	Fourier spectrum
D	0	0	0	0	0	
N	1	0	0	1	0	
V	0	0	1	1	0	6 peaks
AF	0	0	1	1	0	broad ring
VW	1	1	1	1	1	
PL	1	0	1	1	1	

polar model

phase	$P_{\rm g}$	R_ℓ	P_ℓ	remarks
D	0	0	0	
P	1	0	1	
V	0	1	1	4 peaks in Fourier spectrum
VW	1	1	1	1 peak in pdf of polar angles
ZZ	1	1	1	2 peaks in pdf of polar angles
VC	0	1	1	

II. PARAMETERS AND SETUP OF SUPPLEMENTARY MOVIES

The available movies correspond to each and every panel of Fig. 2 of the main text. The last one (`Movie_2.mpg`) shows the run used to produce the space-time plot of Fig. 3 of the main text.

Unless otherwise stated below, all movies have a total duration $\Delta t = 2 \times 10^3$ and start after a transient of 2×10^5. They show a 512×512 area with periodic boundary conditions. Depending on the global density, $10^5 - 10^6$ particles were simulated. For `Movies_a-1, a-2,` $\cdots$ `, g-2.mpg`, please see Fig. 2 of the main text for color tables. For `Movie_2.mpg`, the color table is the one used for local polar order in Fig. 2, not that used in Fig. 3.

- `Movie_a-1.mpg`: local nematic order

  `Movie_a-2.mpg`: density map ($\rho_M = 2$)

  Movies of vortex lattice in nematic model ($m = 2$), where $\rho_0 = 1$, and $\tau = 100$, corresponding to Fig. 2(a).

- `Movie_b-1.mpg`: local nematic order

  `Movie_b-2.mpg`: density map ($\rho_M = 2$)

  Movies of active foam in nematic model ($m = 2$), where $\rho_0 = 1$, and $\tau = 40$, corresponding to Fig. 2(b).

  `Movie_b-1L.mpg`: local nematic order

  `Movie_b-2L.mpg`: density map ($\rho_M = 2$)

  The total duration is 2×10^5. The other conditions are the same as above.

- `Movie_c-1.mpg`: local polar order

  `Movie_c-2.mpg`: density map ($\rho_M = 2$)

  Movies of Vicsek wave in nematic model ($m = 2$), where $\rho_0 = 0.56$, and $\tau = 10$, corresponding to Fig. 2(c).

- `Movie_d-1.mpg`: local polar order

  `Movie_d-2.mpg`: density map ($\rho_M = 2$)

  Movies of polar lane in nematic model ($m = 2$), where $\rho_0 = 1.78$, and $\tau = 20$, corresponding to Fig. 2(d).

- `Movie_e-1.mpg`: local polar order

  `Movie_e-2.mpg`: density map ($\rho_M = 2$)

  Movies of vortex lattice in polar model ($m = 1$), where $\rho_0 = 1$, and $\tau = 100$, corresponding to Fig. 2(e).

- `Movie_f-1.mpg`: local polar order

  `Movie_f-2.mpg`: density map ($\rho_M = 4$)

  Movies of vortex chaos in polar model ($m = 1$), where $\rho_0 = 4.22$, and $\tau = 17.5$, corresponding to Fig. 2(f).

- `Movie_g-1.mpg`: local polar order

  `Movie_g-2.mpg`: density map ($\rho_M = 5$)

  Movies of zigzag in polar model ($m = 1$), where $\rho_0 = 3.16$, and $\tau = 15$, corresponding to Fig. 2(g).

- `Movie_2.mpg`: local polar order

  Movies of polar lane in nematic model ($m = 2$), where $\rho_0 = 1.33$, and $\tau = 20$, corresponding to Fig. 3. Area size is now 1024×256 with periodic boundary conditions.

III. CONTINUOUS THEORY FOR EFFECTIVE VORTEX INTERACTION

A. Polar interaction

Here, we derive the continuous description for the number density of underdamped self-propelled particles in 2D. The particle (or microscopic) equations of motion are of the form

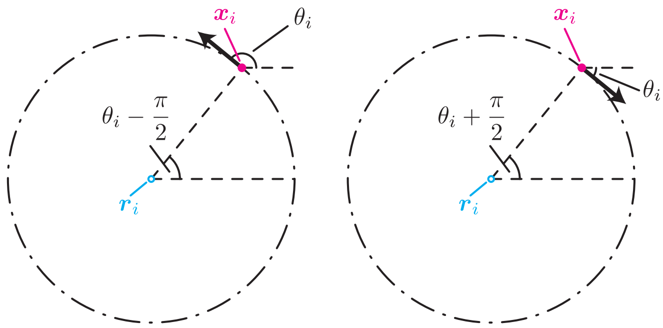

FIG. 1. Definition of $\mathbf{r}_i$. The arrows point in the direction of the motion. The alternate long and short dash lines show the trajectories of the particles. (left) $\epsilon_i = 1$. (right) $\epsilon_i = -1$.

$$\dot{\mathbf{x}}_i = \mathbf{e}_{\theta_i}, \tag{1}$$

$$\dot{\theta}_i = \frac{\alpha}{\mathcal{N}_i} \sum_{|\mathbf{x}_j - \mathbf{x}_i| < 1} \sin\left[\theta_j - \theta_i\right] + \omega_i, \tag{2}$$

where $\mathbf{x}_i$ and θ_i are the position and the direction of motion of particle i, respectively, $\mathbf{e}_{\theta_i} = (\cos\theta_i, \sin\theta_i)$, $\mathcal{N}_i$ is the number of the neighbors of i within unit distance, and ω_i is a telegraphic noise with the value of $\pm\omega_0$ whose average waiting time is τ. Here, we assume for simplicity $\mathcal{N}_i = 1$. In the absence of interactions with other particles, individual particles move along circular arcs of radius $1/\omega_0$. Thus, it is convenient to define the center of the trajectory of particle i as $\mathbf{r}_i$ (Fig. 1), is calculated as

$$\mathbf{r}_i = \mathbf{x}_i - \frac{\epsilon_i}{\omega_0} \mathbf{e}_{\theta_i - \frac{\pi}{2}}, \tag{3}$$

where ϵ_i is the sign of ω_i. Conversely,

$$\mathbf{x}_i = \mathbf{r}_i + \frac{\epsilon_i}{\omega_0} \mathbf{e}_{\theta_i - \frac{\pi}{2}}. \tag{4}$$

Differentiating Eqn. (3) one derives (using $d\epsilon_i(t)/dt = -2\epsilon_i \delta(t - t_s)$)

$$\dot{\mathbf{r}}_i = \frac{2\epsilon_i}{\omega_0} \mathbf{e}_{\theta_i - \frac{\pi}{2}} \delta(t - t_s) - \frac{\alpha\epsilon_i}{\omega_0} \sum_{|\mathbf{x}_j - \mathbf{x}_i| < 1} \sin[\theta_j - \theta_i] \mathbf{e}_{\theta_i}, \tag{5}$$

where t_s is the time when ω_i switches to $-\omega_i$.

In order to derive from Eqs. (2, 5) the continuity equations, we introduce the two number densities of particles at the position $\mathbf{r}$ and orientation θ with respect to the sign of the telegraphic noise ϵ, $f_\epsilon(\mathbf{r}, \theta)$. We define the "collective rotation rates" for each of the states as

$$U_{\pm} = \alpha \int \mathrm{d}\xi \int_{-\pi}^{\pi} \mathrm{d}\phi \; \{\chi(\mathbf{x}^{1}_{\xi,\phi}, \mathbf{x}^{\pm 1}_{\mathbf{r},\theta}) f_{+}(\xi,\phi) + \chi(\mathbf{x}^{-1}_{\xi,\phi}, \mathbf{x}^{\pm 1}_{\mathbf{r},\theta}) f_{-}(\xi,\phi)\} \sin[\phi - \theta]. \tag{6}$$

Then we obtain

$$\begin{aligned} \frac{\partial f_{\pm}(\mathbf{r},\theta)}{\partial t} =& \pm \frac{\partial}{\partial \mathbf{r}} \cdot \mathbf{e}_{\theta} f_{\pm}(\mathbf{r},\theta) \frac{U_{\pm}}{\omega_0} - \frac{\partial}{\partial \theta} f_{\pm}(\mathbf{r},\theta)(\pm\omega_0 + U_{\pm}) \\ &+ \frac{f_{\mp}\left(\mathbf{r} \pm D\mathbf{e}_{\theta-\frac{\pi}{2}}, \theta\right) - f_{\pm}(\mathbf{r},\theta)}{\tau}, \end{aligned} \tag{7}$$

where $D = \frac{2}{\omega_0}$. Here, χ is a local kernel defined as

$$\chi(\mathbf{x}', \mathbf{x}) = \begin{cases} 1, & |\mathbf{x}' - \mathbf{x}| < 1 \\ 0, & |\mathbf{x}' - \mathbf{x}| \geq 1 \end{cases}, \tag{8}$$

and $\mathbf{x}^{\epsilon}_{\mathbf{r},\theta}$ is defined according to Eq. (4). The meaning of the terms in Eq. (7) is in order: the translocation of the centers of rotation (terms $\sim \frac{\partial}{\partial \mathbf{r}} \cdot \mathbf{e}_{\theta} f_{\pm}(\mathbf{r},\theta) U_{\pm}$) and the direction change of the particles ($\sim \frac{\partial}{\partial \theta} f_{\pm}(\mathbf{r},\theta)(\pm\omega_0 + U_{\pm})$) due to alignment interaction with the neighbors, and the jump of the centers when ω_i switches to $-\omega_i$ ($\sim \frac{1}{\tau}\left\{f_{\mp}\left(\mathbf{r} \pm D\mathbf{e}_{\theta-\frac{\pi}{2}}, \theta\right) - f_{\pm}(\mathbf{r},\theta)\right\}$).

In the following, we assume that the distribution of θ is uniform, i.e. functions $f_{\pm}$ do not depend on θ. This assumption is justified below the onset of polar order when θ changes little during one collision. Applying $\frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta$ to both sides of Eqs. (7), the following equations are derived:

$$\begin{aligned} \frac{\partial F_{\pm}}{\partial t} =& - D\alpha \frac{\partial}{\partial \mathbf{r}} \cdot F_{\pm}(\mathbf{r}) \int \mathrm{d}\xi \; \left\{\mathbf{K}^{1}_{\pm 1}(\mathbf{r} - \xi) F_{+}(\xi) + \mathbf{K}^{-1}_{\pm 1}(\mathbf{r} - \xi) F_{-}(\xi)\right\} \\ &+ \frac{1}{\tau}\left\{\frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta \; F_{\mp}\left(\mathbf{r} + D\mathbf{e}_{\theta}\right) - F_{\pm}(\mathbf{r})\right\}, \end{aligned} \tag{9}$$

where $F_{\pm}(\mathbf{r}) = \int_0^{2\pi} \mathrm{d}\theta \; f_{\pm}(\mathbf{r},\theta)/2\pi$, and the corresponding radial kernels $\mathbf{K}^{\mu}_{\epsilon}$ are of the form

$$\mathbf{K}^{\mu}_{\epsilon}(\mathbf{r}) = -\epsilon \int_{-\pi}^{\pi} \mathrm{d}\theta \int_{-\pi}^{\pi} \mathrm{d}\phi \; \chi\left(\mathbf{x}^{\mu}_{\mathbf{0},\phi}, \mathbf{x}^{\epsilon}_{\mathbf{r},\theta}\right) \sin\left[\phi - \theta\right] \frac{\mathbf{e}_{\theta}}{4\pi}. \tag{10}$$

We will show that 4 kernels $\mathbf{K}^{\mu}_{\epsilon}$ are reduced to a single radial kernel $\mathbf{K}$ due to corresponding symmetry conditions, i.e. $\mathbf{K}^{+}_{+} = \mathbf{K}^{-}_{-} = \mathbf{K}$, $\mathbf{K}^{-}_{+} = \mathbf{K}^{+}_{-} = -\mathbf{K}$.

We will use the following change of variables for explicit calculation of the kernels, $\mathit{\Theta}_1 = \frac{\pi}{2} + (\epsilon - \mu)\frac{\pi}{4} + \frac{\phi - \theta}{2}$ and $\mathit{\Theta}_2 = \frac{\pi}{2} + (\epsilon + \mu)\frac{\pi}{4} - \frac{\phi + \theta}{2} + \psi_{\mathbf{r}}$, where $\psi_{\mathbf{r}}$ is the angle between $\mathbf{r}$ and the x-axis (Fig. 2). Then one derives

$$\begin{aligned} \left|\mathbf{x}^{\mu}_{\mathbf{0},\phi} - \mathbf{x}^{\epsilon}_{\mathbf{r},\theta}\right|^2 =& \left(|\mathbf{r}| - \frac{D}{2}\cos\left(\mathit{\Theta}_1 + \mathit{\Theta}_2\right) - \frac{D}{2}\cos\left(\mathit{\Theta}_1 - \mathit{\Theta}_2\right)\right)^2 \\ &+ \left(\frac{D}{2}\sin\left(\mathit{\Theta}_1 + \mathit{\Theta}_2\right) - \frac{D}{2}\sin\left(\mathit{\Theta}_1 - \mathit{\Theta}_2\right)\right)^2 \\ =& |\mathbf{r}|^2 + \frac{D^2}{2} - 2|\mathbf{r}| D \cos\mathit{\Theta}_1 \cos\mathit{\Theta}_2 + \frac{D^2}{2}\cos 2\mathit{\Theta}_1. \end{aligned} \tag{11}$$

When $\left|\mathbf{x}^{\mu}_{\mathbf{0},\phi} - \mathbf{x}^{\epsilon}_{\mathbf{r},\theta}\right| < 1$, $0 \leq |\mathbf{r}| < D + 1$ and

$$\left(\cos\mathit{\Theta}_2 - \frac{|\mathbf{r}|^2 - 1 + D^2\cos^2\mathit{\Theta}_1}{2|\mathbf{r}| D \cos\mathit{\Theta}_1}\right)\cos\mathit{\Theta}_1 > 0. \tag{12}$$

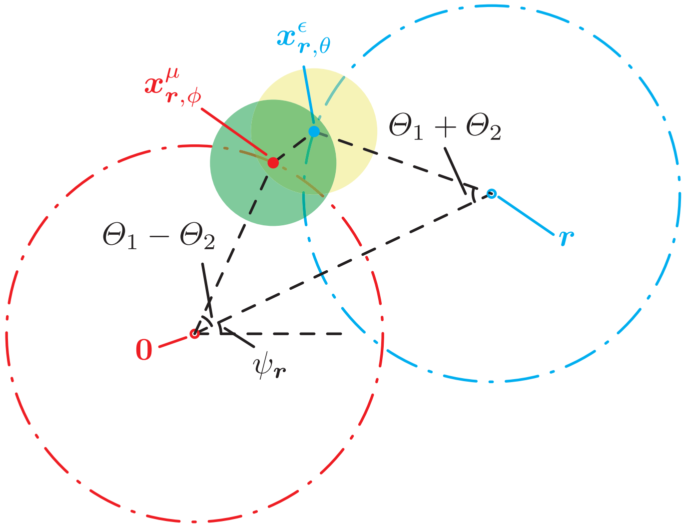

FIG. 2. Schematic image of Θ_1, Θ_2, and $\psi_{\mathbf{r}}$. When θ and ϕ satisfy Eq. (13), the particles are inside the interaction area (the green area and the yellow area) of the other.

Solving inequality (12) we obtain

$$\begin{cases} \left(\cos\Phi<\cos\Theta_2<1, \frac{1-|\mathbf{r}|}{D}<\cos\Theta_1<\frac{|\mathbf{r}|+1}{D}\right), \\ \quad\left(-1<\cos\Theta_2<\cos\Phi, -\frac{|\mathbf{r}|+1}{D}<\cos\Theta_1<-\frac{1-|\mathbf{r}|}{D}\right), \\ \quad\quad\left(-\frac{1-|\mathbf{r}|}{D}<\cos\Theta_1<\frac{1-|\mathbf{r}|}{D}\right) & (0<|\mathbf{r}|<1) \\ \left(\cos\Phi<\cos\Theta_2<1, \frac{|\mathbf{r}|-1}{D}<\cos\Theta_1<\frac{|\mathbf{r}|+1}{D}\right), \\ \quad\left(-1<\cos\Theta_2<\cos\Phi, -\frac{|\mathbf{r}|+1}{D}<\cos\Theta_1<-\frac{|\mathbf{r}|-1}{D}\right) & (1<|\mathbf{r}|<D-1) \\ \left(\cos\Phi<\cos\Theta_2<1, \frac{|\mathbf{r}|-1}{D}<\cos\Theta_1<1\right), \\ \quad\left(-1<\cos\Theta_2<\cos\Phi, -1<\cos\Theta_1<-\frac{|\mathbf{r}|-1}{D}\right) & (D-1<|\mathbf{r}|<D+1) \end{cases}, \tag{13}$$

where $\Phi=\cos^{-1}\frac{|\mathbf{r}|^2-1+D^2\cos^2\Theta_1}{2|\mathbf{r}|D\cos\Theta_1}\quad(0\leq\Phi\leq\pi)$.

Using the above results, one calculates the kernels explicitly:

$$\begin{aligned}\mathbf{K}^{\mu}_{\epsilon}(\mathbf{r})=&-\epsilon\int_{-\pi}^{\pi}\mathrm{d}\Theta_1\int_{-\pi}^{\pi}\mathrm{d}\Theta_2\;\chi\left(\mathbf{x}^{\mu}_{\mathbf{0},\phi},\mathbf{x}^{\epsilon}_{\mathbf{r},\theta}\right)\sin\left[2\Theta_1-(\epsilon-\mu)\frac{\pi}{2}-\pi\right]\frac{\mathbf{e}_{-(\Theta_2+\Theta_1)+\pi+\epsilon\frac{\pi}{2}+\psi_{\mathbf{r}}}}{4\pi}\\ =&\epsilon\mu\mathbf{K}(\mathbf{r}),\end{aligned} \tag{14}$$

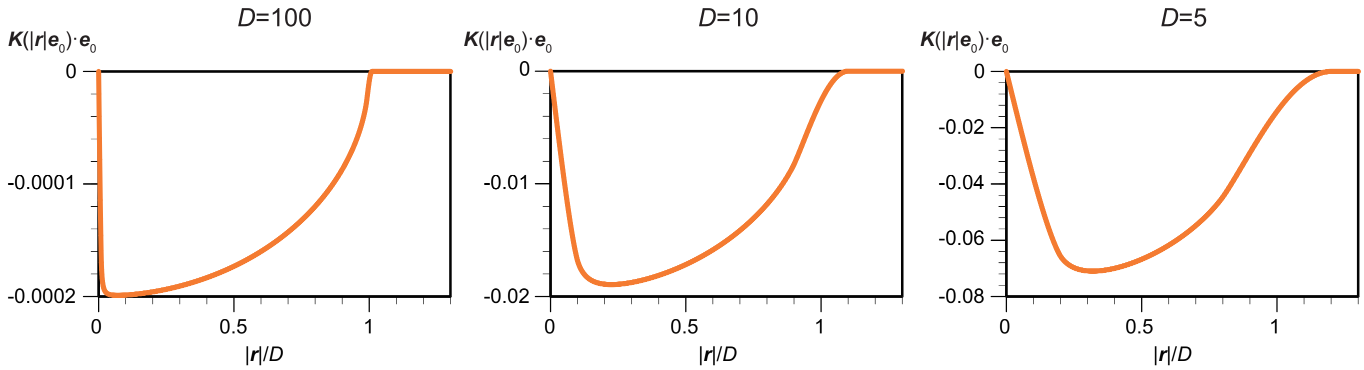

FIG. 3. $\mathbf{K}(\mathbf{r})\cdot\mathbf{e}_{\psi_\mathbf{r}}$ in the Polar case.

where

$$
\begin{aligned}
\mathbf{K}(\mathbf{r}) &= -\frac{\mathbf{e}_{\psi_\mathbf{r}}}{4\pi}\int_{-\pi}^{\pi}\mathrm{d}\Theta_1\int_{-\pi}^{\pi}\mathrm{d}\Theta_2\;\chi\left(\mathbf{x}^1_{\mathbf{0},\phi},\mathbf{x}^1_{\mathbf{r},\theta}\right)\sin 2\Theta_1\cos\left(\frac{\pi}{2}-(\Theta_1+\Theta_2)\right)\\
&= -\frac{\mathbf{e}_{\psi_\mathbf{r}}}{\pi}\begin{cases}
\int_{\cos^{-1}\frac{|\mathbf{r}|+1}{D}}^{\cos^{-1}\frac{1-|\mathbf{r}|}{D}}\mathrm{d}\Theta_1\int_{-\Phi}^{\Phi}\mathrm{d}\Theta_2\;\sin 2\Theta_1\sin(\Theta_1+\Theta_2) & (0\le|\mathbf{r}|<1)\\
\int_{\cos^{-1}\frac{|\mathbf{r}|+1}{D}}^{\cos^{-1}\frac{|\mathbf{r}|-1}{D}}\mathrm{d}\Theta_1\int_{-\Phi}^{\Phi}\mathrm{d}\Theta_2\;\sin 2\Theta_1\sin(\Theta_1+\Theta_2) & (1\le|\mathbf{r}|<D-1)\\
\int_{0}^{\cos^{-1}\frac{|\mathbf{r}|-1}{D}}\mathrm{d}\Theta_1\int_{-\Phi}^{\Phi}\mathrm{d}\Theta_2\;\sin 2\Theta_1\sin(\Theta_1+\Theta_2) & (D-1\le|\mathbf{r}|<D+1)\\
0 & (|\mathbf{r}|\ge D+1)
\end{cases}\\
&= -\frac{2D\mathbf{e}_{\psi_\mathbf{r}}}{\pi|\mathbf{r}|}\begin{cases}
\int_{\frac{1-|\mathbf{r}|}{D}}^{\frac{|\mathbf{r}|+1}{D}}\mathrm{d}s\;\sqrt{(1-s^2)\left\{\left(\frac{|\mathbf{r}|+1}{D}\right)^2-s^2\right\}\left\{s^2-\left(\frac{|\mathbf{r}|-1}{D}\right)^2\right\}} & (0\le|\mathbf{r}|<1)\\
\int_{\frac{|\mathbf{r}|-1}{D}}^{\frac{|\mathbf{r}|+1}{D}}\mathrm{d}s\;\sqrt{(1-s^2)\left\{\left(\frac{|\mathbf{r}|+1}{D}\right)^2-s^2\right\}\left\{s^2-\left(\frac{|\mathbf{r}|-1}{D}\right)^2\right\}} & (1\le|\mathbf{r}|<D-1)\\
\int_{\frac{|\mathbf{r}|-1}{D}}^{1}\mathrm{d}s\;\sqrt{(1-s^2)\left\{\left(\frac{|\mathbf{r}|+1}{D}\right)^2-s^2\right\}\left\{s^2-\left(\frac{|\mathbf{r}|-1}{D}\right)^2\right\}} & (D-1\le|\mathbf{r}|<D+1)\\
0 & (|\mathbf{r}|\ge D+1)
\end{cases}.
\end{aligned}
\tag{15}
$$

The effective force ($\epsilon\mu\mathbf{K}(\mathbf{r})\cdot\mathbf{e}_{\psi_\mathbf{r}}$) is always negative (attractive between vortices of the same sign) or positive (repulsive between vortices of opposite signs) as shown in Fig. 3.

Using the above explicit expressions for the kernels, we can examine the stability of the homogeneous state with the mean number density of particles ρ_0. Linearizing Eqs. (9), and applying the Fourier transformation, we derive equations for the Fourier corresponding coefficients of $F_\pm(\mathbf{r})$, $F^\mathbf{k}_\pm$ as follows:

$$
\begin{aligned}
\frac{\mathrm{d}F^\mathbf{k}_\pm}{\mathrm{d}t} = &\left\{\pi D\alpha\rho_0|\mathbf{k}|\int_0^{D+1}\mathrm{d}r\;J_1(|\mathbf{k}|r)\left(\mathbf{K}(r\mathbf{e}_0)\cdot r\mathbf{e}_0\right)+\frac{J_0(|\mathbf{k}|D)}{\tau}\right\}F^\mathbf{k}_\mp\\
&-\left\{\pi D\alpha\rho_0|\mathbf{k}|\int_0^{D+1}\mathrm{d}r\;J_1(|\mathbf{k}|r)\left(\mathbf{K}(r\mathbf{e}_0)\cdot r\mathbf{e}_0\right)+\frac{1}{\tau}\right\}F^\mathbf{k}_\pm,
\end{aligned}
\tag{16}
$$

where J_n is the Bessel Function of the first kind. Assuming time dependence $F^\mathbf{k}_\pm\sim\exp(\lambda(\mathbf{k})t)$, we derive the corresponding eigenvalues λ for Eq. (16) in the form

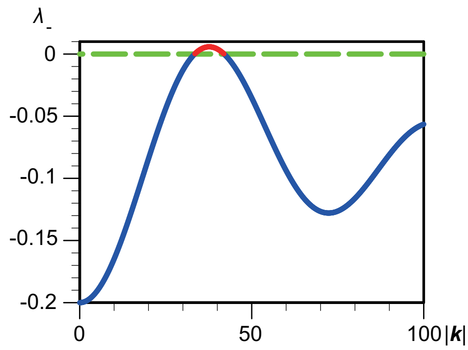

FIG. 4. $\lambda_-(\mathbf{k})$ in the case of the polar interaction. $D = 10$, $\alpha = 0.05$, $\rho_0 = 0.02$, and $\tau = 10$.

$$\lambda_+(\mathbf{k}) = \frac{J_0(|\mathbf{k}|D) - 1}{\tau} \leq 0, \tag{17}$$

for $F_+^{\mathbf{k}} + F_-^{\mathbf{k}}$, and

$$\lambda_-(\mathbf{k}) = -2\pi D\alpha\rho_0|\mathbf{k}| \int_0^{D+1} \mathrm{d}r \; J_1\left(|\mathbf{k}|r\right) \left(\mathbf{K}(r\mathbf{e}_0) \cdot r\mathbf{e}_0\right) + \frac{-J_0(|\mathbf{k}|D) - 1}{\tau}, \tag{18}$$

for $F_+^{\mathbf{k}} - F_-^{\mathbf{k}}$. For $\mathbf{k} \neq 0$ the eigenmode with $\lambda_+ < 0$ is always overdamped and does not affect the linear stability. In contrast, λ_- can become positive when $\rho_0\tau$ is larger than the critical value. There exist $\mathbf{k}$ where $\lambda_-(\mathbf{k}) > 0$ as shown in Fig. 4. Maximum growth rate occurs for finite $\mathbf{k}_m$ that signals the onset of short-wave instability and the onset of periodic structures with the characteristic scale $L_c \sim 1/|\mathbf{k}_m|$.

B. Nematic interaction

In this section, we examine the continuous description for the case of nematic interaction. With the same manner in the polar case, the following continuous equations are obtained:

$$\begin{aligned} \frac{\partial F_\pm}{\partial t} =& -D\alpha \frac{\partial}{\partial \mathbf{r}} \cdot F_\pm(\mathbf{r}) \int \mathrm{d}\xi \; \mathbf{K}(\mathbf{r} - \xi) \left\{F_+(\xi) + F_-(\xi)\right\} \\ &+ \frac{1}{\tau} \left\{ \frac{1}{2\pi} \int_0^{2\pi} \mathrm{d}\theta \; F_\mp\left(\mathbf{r} + D\mathbf{e}_\theta\right) - F_\pm(\mathbf{r}) \right\}, \end{aligned} \tag{19}$$

where radial kernel $\mathbf{K}$ is of the from

$$\mathbf{K}(\mathbf{r}) = \frac{4D\mathbf{e}_{\psi_\mathbf{r}}}{\pi|\mathbf{r}|} \begin{cases} \int_{\frac{1-|\mathbf{r}|}{D}}^{\frac{|\mathbf{r}|+1}{D}} \mathrm{d}s \;\; (2s^2 - 1) \sqrt{(1-s^2) \left\{ \left(\frac{|\mathbf{r}|+1}{D}\right)^2 - s^2 \right\} \left\{ s^2 - \left(\frac{|\mathbf{r}|-1}{D}\right)^2 \right\}} & (0 \leq |\mathbf{r}| < 1) \\ \int_{\frac{|\mathbf{r}|-1}{D}}^{\frac{|\mathbf{r}|+1}{D}} \mathrm{d}s \;\; (2s^2 - 1) \sqrt{(1-s^2) \left\{ \left(\frac{|\mathbf{r}|+1}{D}\right)^2 - s^2 \right\} \left\{ s^2 - \left(\frac{|\mathbf{r}|-1}{D}\right)^2 \right\}} & (1 \leq |\mathbf{r}| < D-1) \\ \int_{\frac{|\mathbf{r}|-1}{D}}^{1} \mathrm{d}s \;\; (2s^2 - 1) \sqrt{(1-s^2) \left\{ \left(\frac{|\mathbf{r}|+1}{D}\right)^2 - s^2 \right\} \left\{ s^2 - \left(\frac{|\mathbf{r}|-1}{D}\right)^2 \right\}} & (D-1 \leq |\mathbf{r}| < D+1) \\ 0 & (\mathbf{r} \geq D+1) \end{cases} \tag{20}$$

The effective force $(\mathbf{K}(\mathbf{r}) \cdot \mathbf{e}_{\psi_\mathbf{r}})$ is shown in Fig. 5. The interaction has a short-range attractive part and a medium range repulsive part.

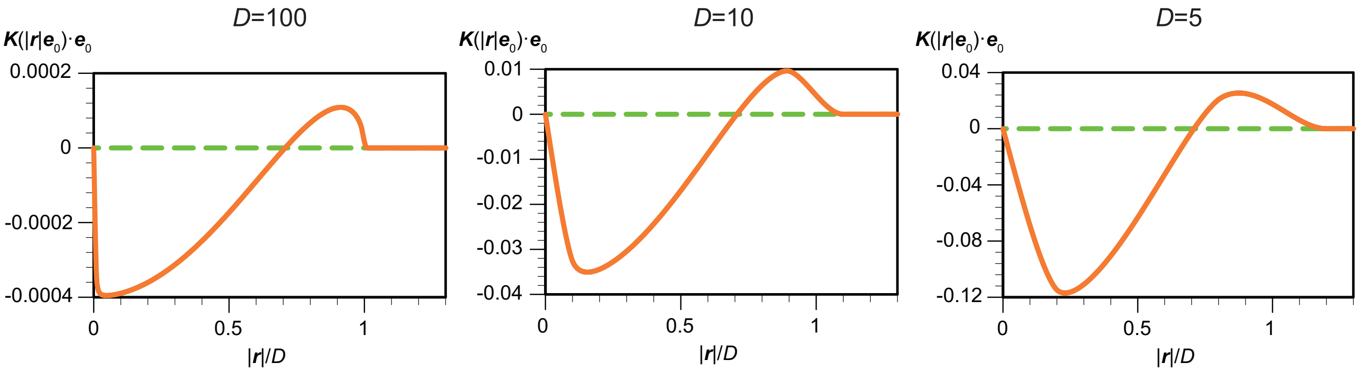

FIG. 5. $\mathbf{K}(\mathbf{r}) \cdot \mathbf{e}_{\psi_\mathbf{r}}$ in the nematic case.

Correspondingly, linearized equations for the Fourier coefficients of $F_\pm$ are of the form

$$\frac{\mathrm{d}F_\pm^\mathbf{k}}{\mathrm{d}t} = \left\{ -\pi D\alpha\rho_0|\mathbf{k}| \int_0^{D+1} \mathrm{d}r\ J_1\left(|\mathbf{k}|r\right)\left(\mathbf{K}(r\mathbf{e}_0)\cdot r\mathbf{e}_0\right) + \frac{J_0(|\mathbf{k}|D)}{\tau} \right\} F_\mp^\mathbf{k} - \left\{ \pi D\alpha\rho_0|\mathbf{k}| \int_0^{D+1} \mathrm{d}r\ J_1\left(|\mathbf{k}|r\right)\left(\mathbf{K}(r\mathbf{e}_0)\cdot r\mathbf{e}_0\right) + \frac{1}{\tau} \right\} F_\pm^\mathbf{k}, \tag{21}$$

The eigenvalues $\lambda(\mathbf{k})$ of Eqs. (21) are of the from

$$\lambda_+(\mathbf{k}) = -2\pi D\alpha\rho_0|\mathbf{k}| \int_0^{D+1} \mathrm{d}r\ J_1\left(|\mathbf{k}|r\right)\left(\mathbf{K}(r\mathbf{e}_0)\cdot r\mathbf{e}_0\right) + \frac{J_0(|\mathbf{k}|D) - 1}{\tau} \tag{22}$$

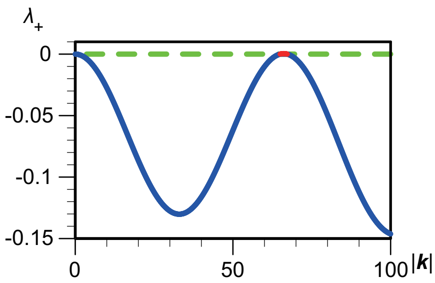

FIG. 6. $\lambda_+(\mathbf{k})$ in the case of the nematic interaction. $D = 10$, $\alpha = 0.05$, $\rho_0 = 0.02$, and $\tau = 8.5$.

for $F_+^\mathbf{k} + F_-^\mathbf{k}$, and

$$\lambda_-(\mathbf{k}) = -\frac{J_0(|\mathbf{k}|D) + 1}{\tau} < 0 \tag{23}$$

for $F_+^\mathbf{k} - F_-^\mathbf{k}$. In contrast to the polar case, in the nematic case the eigenvalue λ_- is always negative. When $\rho_0\tau$ is larger than the critical value, there exist $\mathbf{k}$ where $\lambda_+(\mathbf{k}) > 0$ as shown in Fig. 6. Again, the bifurcation scenario signals the onset short-wave instability with the characteristic scale $L_c = 1/|\mathbf{k}_m|$.